- Passive element – The element which either store the energy or dissipate the energy is called passive element.
  e.g. resistor, inductor, Capacitor

- Unilateral element – The element which allow current to flow in one dir^n is called unilateral element.
  e.g. Diode

- Bilateral element – The element which allow current to flow in both the dir^n is called bilateral element.
  e.g. Resistor

- Linear element – The element in which Voltage and Current are related to a linear equation.
  The element which obey ohm's law is called linear element.
  e.g. Resistor, inductor, Capacitor

- Non-linear element – The element in which V and I are related through a non linear eq^n is called non-linear element.
  The element which does not follow ohm's law
  e.g. Diode

- Lumped Network : A lumped network is a network in which all the network elements are physically Separable.
  e.g. Most of the network are lumped network.

- Distributed network : The network in which all element are not physically separable.
  e.g. Transmission line

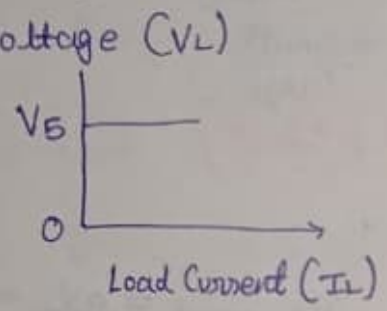
Types of Energy Source

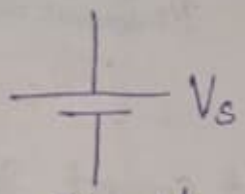
- Voltage Source
  - Ideal / Const. Voltage Source
  - Real / Practical Voltage Source
- Current Source
  - Ideal / Const. Current Source
  - Real practical Current Source

- **Ideal / Constant Voltage Source :**

The Source which provide Constant Voltage to the load irrespective of load Connected across it.

Ideal Voltage Source is not practically possible.



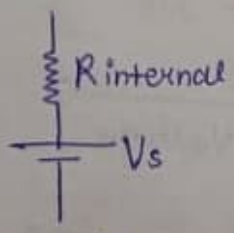


Circuit

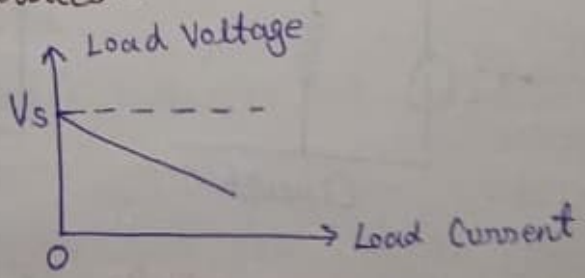
- **VI characteristic of Ideal Voltage Source :**

$$R_{internal} = \frac{\Delta V}{\Delta I} = \frac{Zero}{Large} = Zero$$

② Real / Practical Voltage Sources –



Circuit

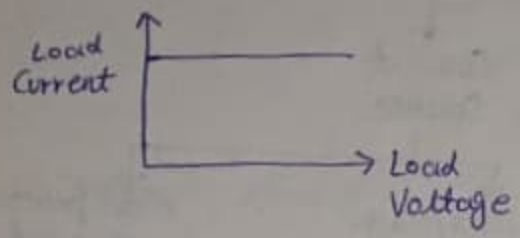


VI characteristics of real voltage Source

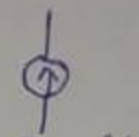
$$R_{internal} = \frac{\Delta V}{\Delta I} = \frac{Small}{Large}$$

$$\boxed{R_{internal} = Small}$$

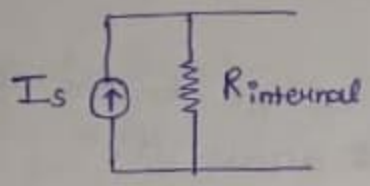
③ Ideal / Constant Current Source :

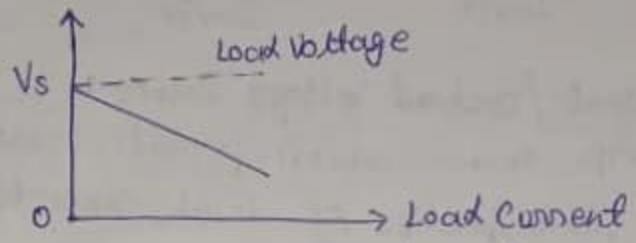


$R_{internal} = \frac{\Delta V}{\Delta I} = \infty$



④ Real / Current Source :



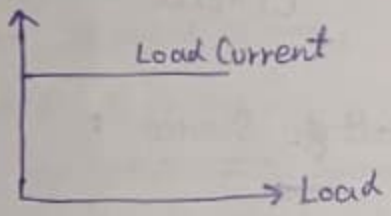


VI characteristic of real Voltage Source

$R_{internal} = \frac{\Delta V}{\Delta I} = \frac{Small}{large}$

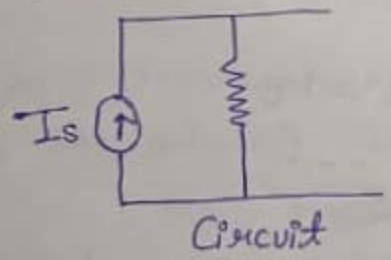
$\boxed{R_{internal} = Small}$

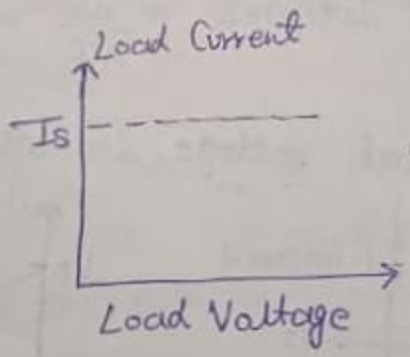
⑤ Ideal / Constant Current Source :



$R_{internal} = \frac{\Delta V}{\Delta I} = \infty$

⑥ Real / Current Source :





$R_{internal} = \frac{\Delta V}{\Delta I}$

$= \frac{Large}{Small}$

$\Rightarrow$ Large

* Never apply KVL to Current Source.

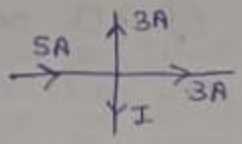
Kirchhoff's Law -

① KCL - The sum of incoming current at a node or junction must be equal to some of outgoing current.

$$5 = 3 + 3 + I$$

$$5 - 6 = I$$

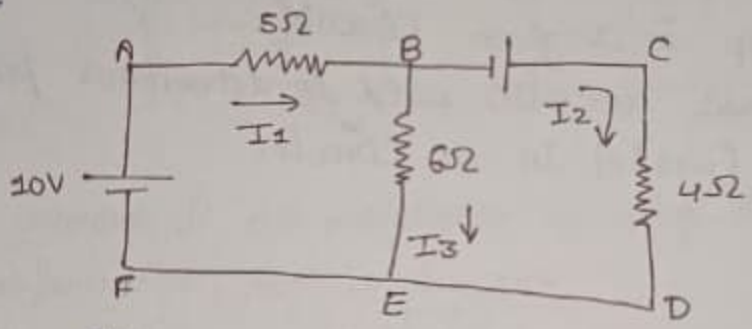
$$\boxed{I = -1A}$$



KCL is based on Conservation of charge because at node there is no accumulation of charge.

② KVL → Is based on Conservation of energy because inclosed path net work done is zero.

According to KVL the algebric sum of all the voltage drops and battery in a closed path must be equal to zero.



$$5I_1 + 6I_3 - 10 = 0$$

Apply KVL in Mesh ABEFA

$$5I_1 + 6I_3 - 10 = 0$$

Apply KVL in mesh BCDEB

$$-20 + 4I_2 - 6I_3 = 0$$

Advantages of KVL and KCL

① KCL and KVL are applicable both for DC and AC Circuit.

② KCL and KVL are applicable both for linear and non-linear Circuit.

Disadvantage of KCL and KVL –

① The loss of KCL and KVL are not suitable for AC Circuit of high frequency signal (we neglect the emf of the circuit therefore it frequency is high).

② KCL and KVL is work under the assumption that there is no fluctuating magnetic field.

③ KCL and KVL is not applicable to distributed network.

→ Application of KVL and KCL :

1) KCL and KVL is used to find unknown current and Voltage drop in a given Circuit.
KCL and KVL are also used to determines power dissipation (waste) in a Circuit.

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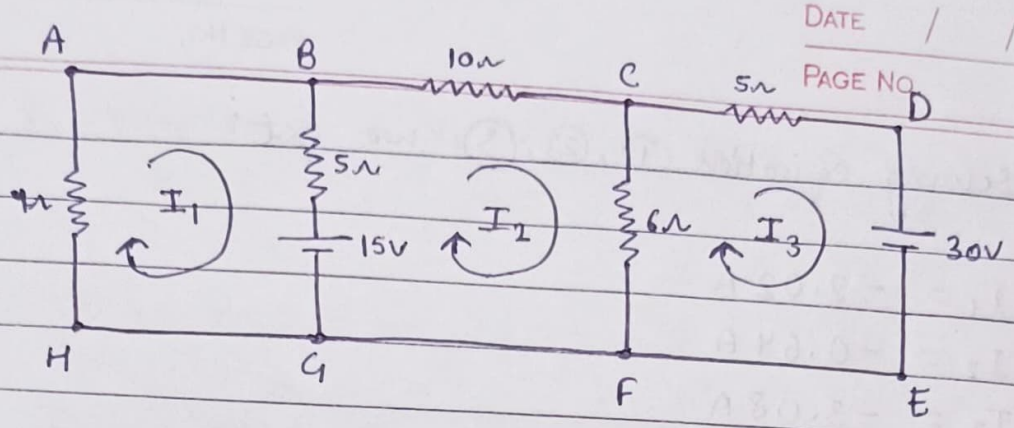
Q2

(a)



→ Applying KVL at Mesh 1 (ABGHA):

$$4I_1 + 5(I_1 - I_2) + 15 = 0$$

$$4I_1 + 5I_1 - 5I_2 + 15 = 0$$

$$\boxed{9I_1 - 5I_2 + 0I_3 = -15} \quad -(1)$$

→ Applying KVL at Mesh 2 (BCFGB):

$$5(I_2 - I_1) + 10I_2 + 6(I_2 - I_3) - 15 = 0$$

$$5I_2 - 5I_1 + 10I_2 + 6I_2 - 6I_3 = 15$$

$$\boxed{-5I_1 + 21I_2 - 6I_3 = 15} \quad -(2)$$

→ Applying KVL at Mesh 3 (CDEFC):

$$6(I_3 - I_2) + 5I_3 + 30 = 0$$

$$6I_3 - 6I_2 + 5I_3 = -30$$

$$\boxed{0I_1 - 6I_2 + 11I_3 = -30} \quad -(3)$$

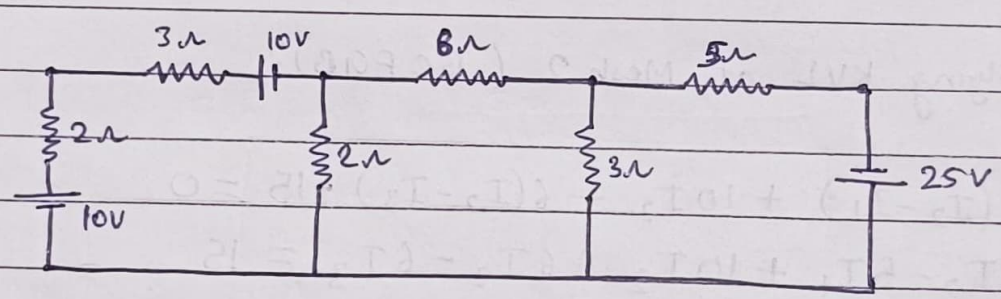
Solving equation ①, ②, ③ we get:

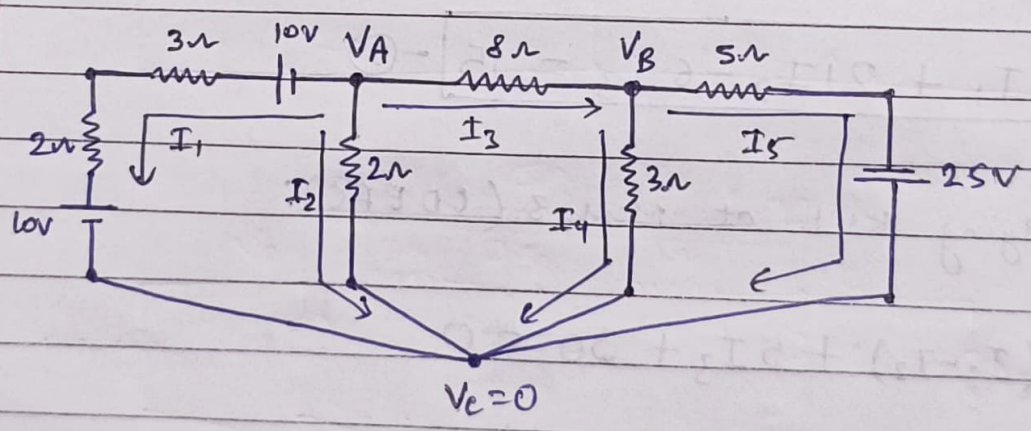
$I_1 = -2.02\,A$
$I_2 = -0.64\,A$
$I_3 = -3.08\,A$

→ Current in all resistances

across $4\Omega \rightarrow I_1 = -2.02\,A$
$10\Omega \rightarrow I_2 = -0.64\,A$
$5\Omega \rightarrow I_2 - I_1 = 1.38\,A$
$6\Omega \rightarrow I_2 - I_3 = 2.44\,A$
$5\Omega \rightarrow I_3 = -3.08\,A$

(a)





Applying KCL at node A:

$I_1 + I_2 + I_3 = 0$

$$\frac{V_A - V_C + 10 - 10}{5} + \frac{V_A - V_C}{2} + \frac{V_A - V_B}{8} = 0$$

$$\frac{V_A}{5} + \frac{V_A}{2} + \frac{V_A - V_B}{8} = 0$$

$$\frac{8V_A + 20V_A + 5V_A - 5V_B}{40} = 0$$

$$\boxed{33V_A - 5V_B = 0} \quad - (1)$$

→ Applying KCL at node B:

$$I_3 = I_4 + I_5$$

$$\frac{V_A - V_B}{8} = \frac{V_B - V_C}{3} + \frac{V_B - V_C + 25}{5}$$

$$\frac{V_A - V_B}{8} = \frac{5V_B - 5V_C + 3V_B - 3V_C + 75}{15}$$

$$15V_A - 15V_B = 64V_B + 600$$

$$\boxed{15V_A - 79V_B = 600} \quad - (2)$$

solving equation (1) and (2)

$V_A = -1.18\,V$

$V_B = -7.81\,V$

Current in all resistances:

Across $3\Omega \rightarrow \frac{V_A - V_C}{5} = -0.23\,A$

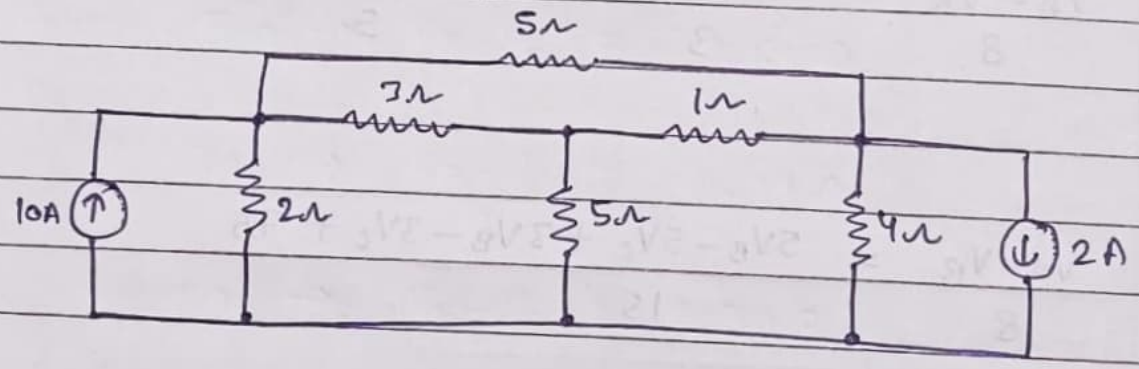
$2\Omega \rightarrow \frac{V_A - V_C}{5} = -0.23\,A$

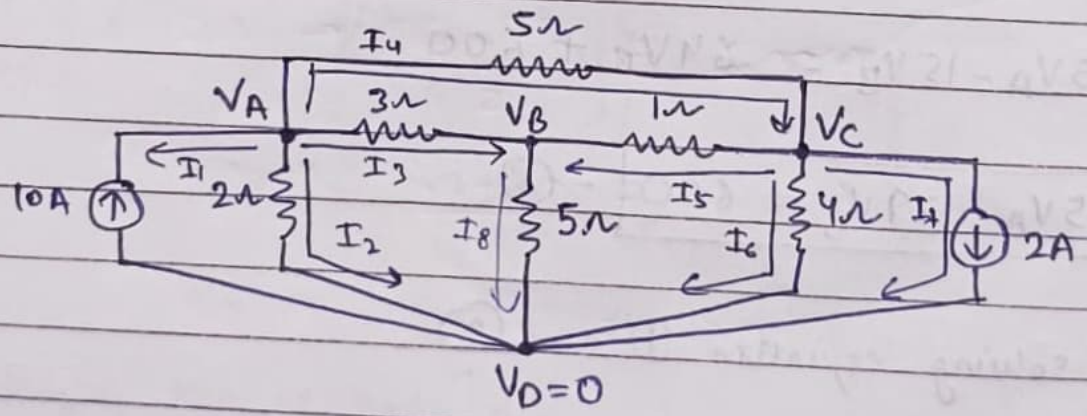
$2\Omega \rightarrow \frac{V_A - V_C}{2} = -0.59\,A$

$8\Omega \rightarrow \frac{V_A - V_B}{8} = 0.82\,A$

$3\Omega \rightarrow \frac{V_B - V_C}{2} = -2.60\,A$

$5\Omega \rightarrow \frac{V_B - V_C + 2\Omega}{5} = 3.43\,A$





→ Apply KCL at node A

$$I_1 + I_2 + I_3 + I_4 = 0$$

$$-10 + \frac{V_A - V_D}{2} + \frac{V_A - V_B}{3} + \frac{V_A - V_C}{5} = 0$$

$$\frac{-300 + 15V_A + 10V_A - 10V_B + 6V_A - 6V_C}{30} = 0$$

$$\boxed{31V_A - 10V_B - 6V_C = 300} \quad - (1)$$

→ Apply KCL at node B

$$I_3 + I_5 = I_8$$

$$\frac{V_A - V_B}{3} + \frac{V_C - V_B}{1} = \frac{V_B - V_D}{5}$$

$$\frac{V_A - V_B + 3V_C - 3V_B}{3} = \frac{V_B}{5}$$

$$5V_A - 20V_B + 15V_C = 3V_B$$

$$\boxed{5V_A - 23V_B + 15V_C = 0} \quad - (2)$$

→ Apply KCL at node C

$$I_4 = I_5 + I_6 + I_7$$

$$\frac{V_A - V_C}{5} = \frac{V_C - V_B}{1} + \frac{V_C - V_D}{4} + 2$$

$$\frac{V_A - V_C}{5} = \frac{4V_C - 4V_B + V_C + 8}{4}$$

$$4V_A - 4V_C = 20V_C - 20V_B + 5V_C + 40$$

$$\boxed{4V_A + 20V_B - 29V_C = 40} \quad -③$$

Solving equation ①, ②, ③ we get:

$V_A = 12.05\,V$
$V_B = 5.10\,V$
$V_C = 3.80\,V$

Current in all resistances

Across $5\Omega \rightarrow \frac{V_A - V_C}{5} = 1.65\,A$

$3\Omega \rightarrow \frac{V_A - V_B}{3} = 2.31\,A$

$1\Omega \rightarrow V_B - V_C = 1.30\,A$

$2\Omega \rightarrow \frac{V_A - V_D}{2} = 6.02\,A$

$5\Omega \rightarrow \frac{V_B - V_D}{5} = 1.02\,A$

$4\Omega \rightarrow \frac{V_C - V_D}{4} = 0.95\,A$